



The basics of mechanical ventilation in adults

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Abstract: Mechanical ventilation (MV) is a complex, labor-intensive, often life-saving process that requires a knowledgeable team to manage. Nurses play a critical role in caring for adults receiving MV. Therefore, they need to know about all aspects of providing ventilatory support. This article reviews the basics of MV, describes how mechanical ventilators operate, and discusses mechanical ventilator management and the practical aspects of caring for adults requiring MV.

Keywords: breathing, COVID-19, mechanical ventilation, respiratory failure

During the height of the COVID-19 pandemic, staffing challenges required nurses to care for patients on mechanical ventilation (MV). However, many did not have experience with MV, as these nurses did not typically work with this intervention before the pandemic. This article reviews the basics of MV, provides a better understanding of how mechanical ventilators operate, and discusses mechanical ventilator management and the practical aspects of caring for adults requiring MV.

Indications for MV

MV takes over a major portion of the work of spontaneous breathing and

provides time for recovery. A patient needs MV when they are either in or are moving into respiratory failure.^{1,2} For example, MV would be appropriate for survivors of a traumatic event that affected cardiopulmonary function, such as a motor vehicle crash, gunshot wound, house fire, or building collapse. Others may need MV after a drug overdose; postsurgical recovery; chronic obstructive pulmonary disease (COPD) exacerbation; complications of a pulmonary infection; sepsis; or failure of other organ systems, such as heart or renal failure, postcardiac arrest, and stroke.

MV provides supplemental oxygen to increase tissue oxygenation

and supports ventilation to remove carbon dioxide, a waste product of metabolism. In addition, MV can help recruit collapsed alveoli and keep them open, increasing the efficiency of gas exchange in the lungs (specifically oxygen and carbon dioxide).

Patient interface and ventilator circuit

MV is either noninvasive or invasive. The former usually comprises an oro-nasal mask secured with straps or headgear. The latter features an endotracheal tube (ETT) or tracheostomy tube.^{1,2}

Noninvasive ventilation (NIV) is often tried in patients who may only need ventilatory support for a short time. If successful, the patient may be ventilated by mask for 24 to 48 hours before weaning or discontinuation. If NIV is unsuccessful due to lack of toleration, ineffectiveness, or if MV is expected to be necessary for a more extended period, the patient will be intubated with an ETT. If the patient needs MV for a prolonged period, the ETT may be replaced by a tracheostomy tube. The timing of this change varies but may occur after 5 to 10 days based on the expectation that venti-

latory support is needed for several more days or longer.

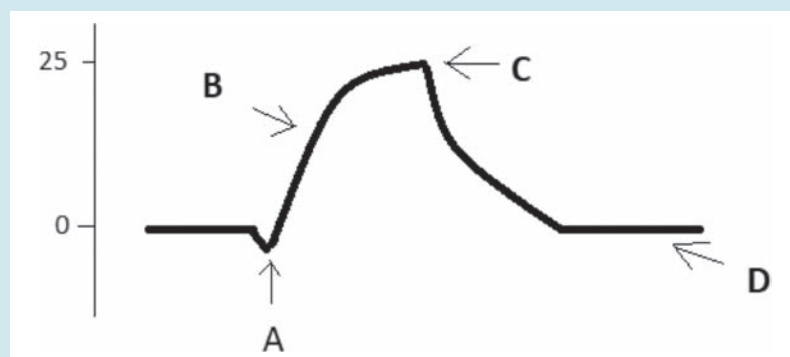
The patient is supplied air from the ventilator through a piece of flexible tubing from the ventilator's inspiratory port. Inspiratory gas exits the ventilator and passes through a heated humidifier mounted on the side or through a heat and moisture exchanger placed at the airway connection. NIV uses only the heated humidifier. After this, the inspired gas is delivered to the interface and into the patient's airway. Exhaled gas from the patient travels back to the ventilator through another flexible tube and goes through a high-efficiency particulate absorbing (HEPA) filter before the gas enters the ventilator. This filter protects the ventilator from internal contamination and the environment from possible infectious agents.

Abbreviations

ABG	Arterial blood gas
CPAP	Continuous positive airway pressure
EBM	Evidence-based medicine
ETT	Endotracheal tube
FIO ₂	Fraction of inspired oxygen
FRC	Functional residual capacity
HAI	Healthcare-associated infections
HEPA	High-efficiency particulate absorbing (filter)
IBW	Ideal body weight
IMV	Intermittent mandatory ventilation
MV	Mechanical ventilation
NIV	Noninvasive ventilation
PC	Pressure controlled
PC-CMV	Pressure-controlled continuous mandatory ventilation
PC-IMV	Pressure-controlled intermittent mandatory ventilation
PEEP	Positive end-expiratory pressure
PRVC	Pressure-regulated volume control
PS	Pressure support
SAT	Spontaneous awakening trial
SBT	Spontaneous breathing trial
VAP	Ventilator-associated pneumonia
VC	Volume controlled
VC-CMV	Volume-controlled continuous mandatory ventilation
VC-IMV	Volume-controlled intermittent mandatory ventilation
WOB	Work of breathing

The ventilator circuit (inspiratory/expiratory flexible tubes) is usually about 5 or 6 feet long. The ventilator is connected to an electrical outlet with an emergency power supply, oxygen, and portable or stationary compressed air outlets.

Four phases of the mechanical ventilator breath (cmH₂O)



A - Trigger phase. Patient effort drops pressure of flow and a breath begins. B - Inspiratory phase. Flow is delivered and pressure builds in the patient-ventilator system as the breath is given. C - Cycle phase. A preset, desired volume is reached and inspiration stops, expiration begins. D - Expiratory phase. The patient-ventilator system returns to baseline. Note the peak pressure at the end of the inspiratory phase is 25 cmH₂O and the baseline is 0 cmH₂O.

The mechanical ventilator breath

Patients receiving MV are ventilated with a positive pressure breath. This means that the ventilator pushes the air into the patient during inspiration and the patient passively exhales due to chest wall and lung recoil. A ventilator breath has four phases or variables: trigger, inspiratory, cycle, and expiratory (see *Four phases of the mechanical ventilator breath*).¹

The *trigger phase* relates to how a ventilator breath is initiated. The trigger can be based on patient effort

(generated either by a drop in the circuit pressure or a drop in a constant flow coming through the ventilator circuit) or triggered by elapsed time without patient effort. The ventilator detects the pressure or flow drop generated by the patient's inspiratory effort, which must reach a threshold to trigger a breath. The threshold is established by the "sensitivity" setting that the respiratory therapist sets in the initial patient-ventilator system start-up. The flow-triggering approach (with time triggering in the background) is used for most patients requiring MV as it is thought to reduce the work of breathing (WOB).³

The time trigger depends on the breath rate set on the ventilator. For example, with a rate or frequency (f) set at 10 breaths/minute (b/min), the ventilator will give a breath every 6 seconds using the time trigger if the patient is not triggering the ventilator with spontaneous efforts. A rate of 12 b/min would result in a breath every 5 seconds, a rate of 15 b/min would result in a breath every 4 seconds, and 20 b/min would result in a breath every 3 seconds. These are the minimum b/min, though the patient can trigger more breaths above this minimum.

Sensitivity must be set so the ventilator will sense the patient's effort to take a breath and initiate the breath. If not sensitive enough, the patient must work harder to get a breath. On the other hand, if too sensitive, the ventilator can auto-

Definition of terms

Tidal volume (Vt)	the amount of air inhaled/exhaled in a normal breath (units = mL). This may be the patient's spontaneous Vt or may be a setting on the ventilator to establish the volume of a mandatory breath
Positive end-expiratory pressure (PEEP)	the positive pressure remaining in the lungs after exhalation, established by the operator (units = cm H ₂ O)
Continuous positive airway pressure (CPAP)	similar to PEEP but this term is used when the patient is breathing spontaneously with no set rate that imposes a breath (units = cm H ₂ O)
Functional residual capacity (FRC)	the volume of air in the lungs at the end of resting exhalation (units = liters)
Minute ventilation	the volume of air provided to the patient over 1 minute (units = liters/minute)
Residual capacity	the volume of air in the lungs at the end of resting exhalation

trigger and cause hyperventilation by giving breaths when the patient did not trigger them.

The *inspiratory phase* relates to the gas flow being pushed into the lungs. Peak inspiratory flow (how fast the ventilator pushes air through the circuit and into the patient) and the ventilator's inspiratory flow pattern (square wave or constant flow versus decelerating or tapering flow patterns) are part of the ventilator settings and selected at the initial set-up. Most commonly, a decelerating flow pattern is used. For adults, the flow is set to provide an inspiratory time of about 1 second (general range of 0.8 to 1.2 seconds). An initial flow setting is usually about 60 L/min (range of 40 to 80 L/min).^{4,5}

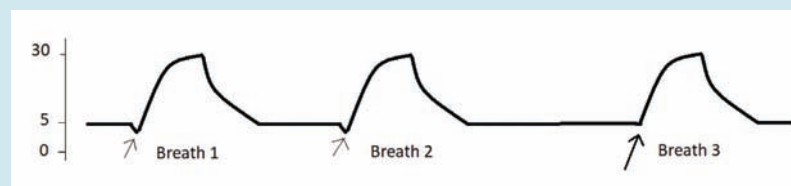
The *cycle phase* relates to how inspiration ends and expiration begins and can be adjusted by certain ventilator settings. For example, the cycle phase is often based on a preset (or desired) tidal volume (Vt) ("volume-cycled") by delivering flow over a period of time to reach the set volume, (see *Definition of terms*). Another approach is based on a preset inspiratory pressure ("pressure-cycled"), using a preset or targeted pressure.

The *expiratory phase* relates to the passive exhalation of the breath, the pause between breaths, and the baseline pressure. Baseline pressure may be zero (meaning atmospheric pressure with no added pressure held in the lungs) and can be adjusted by adding positive end-expiratory pressure (PEEP) or continuous positive airway pressure (CPAP). Many institutions use a PEEP of 5 cm H₂O as a standard initial setting and titrate the setting to higher levels if oxygenation is an issue (see *Triggering and PEEP*).

Ventilator modes and settings

Many adult patients receiving MV are ventilated with a volume-controlled (VC) mode or volume-limited assist

Triggering and PEEP



Three breaths are displayed along with a PEEP of 5 cm H₂O as noted by the baseline elevated above 0 cm H₂O. The first two breaths reflect the patient effort triggering each breath (these could be pressure-triggered or flow-triggered). Note the delay before breath 3. There is no patient effort to initiate breath 3, as seen in the first two breaths. This breath is time-triggered.

control (A/C) to deliver a preset Vt for each mandatory breath (see *Types of breaths during mechanical ventilation*). Others will be set up on a pressure-controlled (PC) mode to deliver a preset pressure for each mandatory breath. This is also called pressure-limited A/C. Either of these approaches can also be set up as continuous mandatory ventilation (CMV) or intermittent mandatory ventilation (IMV). These set-up times are called synchronized IMV. This establishes four major modes based on these combinations: VC-CMV, PC-CMV, VC-IMV, and PC-IMV.⁵ NIV is most often provided by PC-IMV. Also, note that CMV is often referred to as assist/control or A/C.⁶ For many patients, according to this author's experiences, the initial mode of MV is VC-CMV.

Continuous mandatory ventilation

For the VC-CMV mode, the breath, triggered by either the patient or time interval, will have a set Vt. For the PC-CMV mode, the breath, whether triggered by the patient or time, will have a set pressure. It is important to note that each delivered breath will have the same Vt or pressure in both CMV modes.

The CMV modes allow for the most support and relieve the patient of almost all of the WOB. If a patient receiving the CMV mode becomes apneic, all breaths will be given based on the frequency or rate setting and will be time-triggered.



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Intermittent mandatory ventilation

In the IMV modes with either a volume approach (VC-IMV) or pressure approach (PC-IMV), the patient will receive the minimum breaths as determined by the rate or frequency setting and can take additional spontaneous breaths in between the mandatory breaths. In VC-IMV, the mandatory ventilator breaths will reach the set Vt, and the in-between breaths will be defined by the patient's effort, with the volume delivered matched to their effort. In PC-IMV, the mandatory ventilator breaths will reach the preset pressure and the in-between breaths will be the same as that in VC-IMV.

Both IMV modes place more of the WOB on the patient as they take a spontaneous breath between the mandatory breaths. Generally, the rate or frequency used for the mandatory breaths will be lower in IMV than in CMV since the patient is al-

lowed to breathe spontaneously between the mandatory breaths. If a patient receiving the IMV mode becomes apneic, all breaths will be given based on the frequency or rate setting and will be time-triggered. If the frequency is set at a low setting, the patient who is apneic may be hypoventilated.

Positive end-expiratory pressure

PEEP can be added to any of the four major modes (VC-CMV, PC-CMV, VC-IMV, PC-IMV). PEEP allows the patient to exhale passively but will stop a complete exhalation, resulting in a positive baseline pressure. This added pressure increases the lung's functional residual capacity with the goals of keeping the alveoli open, preventing alveolar collapse, and recruiting those alveoli that are being influenced to collapse by outside forces such as a pneumothorax, airway obstruction that is reducing or stopping air from entering the alveoli, or alveoli that are filling with fluid due to infection, pulmonary edema, or inflammation.

The desired effects of adding PEEP are to support and enhance functioning alveoli and improve oxygenation. PEEP settings are balanced with the oxygen setting (FIO₂). Adding PEEP can help reduce the FIO₂, which helps avoid oxygen toxicity. How-

Types of breaths during mechanical ventilation

Mandatory: Breaths are initiated by the ventilator and the ventilator performs the work of inspiration during those breaths.

Assisted: Breaths are initiated by the patient, but the ventilator performs at least some of the work of inspiration for those patient-initiated breaths. This includes the addition of PS where breaths are initiated by the patient and additional inspiratory flow is delivered to cause the inspiratory pressure to reach a set point (established by the operator).

Spontaneous: Breaths are initiated by the patient and the patient performs the entire work of inspiration for those patient-initiated breaths supported.

ever, if too much PEEP is added, too much pressure and volume remain in the lung at baseline, which can decrease stroke volume, cardiac output, and BP. The decrease in cardiac output and stroke volume is due to the increased intrathoracic pressure, which tends to impede venous return to the right heart. This negative effect on the cardiovascular system may also appear without added PEEP due to using positive pressure to ventilate the patient.^{1,6}

Pressure support

Pressure support (PS) can be added to the two IMV modes but not to the two CMV modes. PS occurs only with spontaneous breaths (those breaths that occur in between the mandatory breaths). In PS, a preset pressure is added to the patient's spontaneous breath and provides inspiratory flow to boost the breath to the preset inspiratory pressure. The added flow increases the volume of the spontaneous breath and reduces the WOB during the spontaneous breaths. The diameter of the artificial airway (ETT or tracheostomy tube) can impact the patient's WOB (for example, resistance to airflow



The decision to initiate MV is guided by the patient's clinical status or potential for serious cardiopulmonary deterioration and is based on several factors.

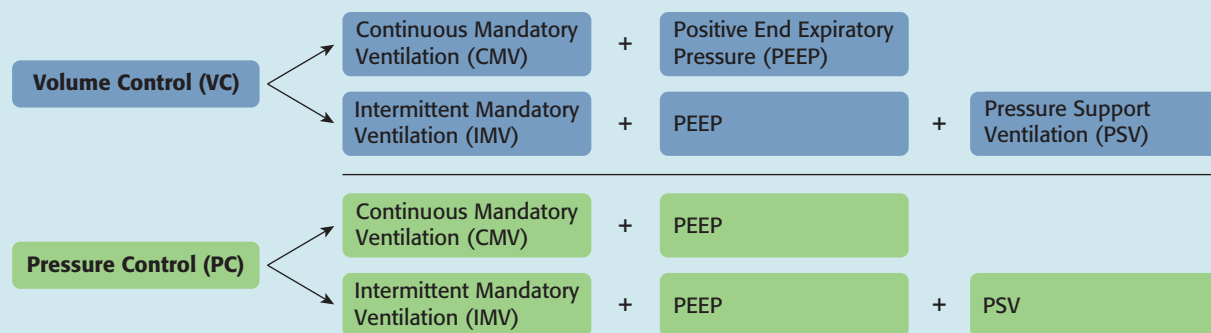
increases as the diameter of the airway decreases, which increases WOB); PS helps relieve this added work. Also, using the largest acceptable-sized artificial airway is recommended to reduce resistance.

Increasing the PS setting will increase the spontaneous Vt and minute ventilation. Minute ventilation is the volume of air (liters/minute) provided to the patient over 1 minute. To calculate the minute ventilation multiply the frequency by the set Vt. PS may be added at a low level to overcome artificial airway resistance. The order for the mode may be VC-IMV + PS or PC-IMV + PS. Note that PEEP can also be added to all possible modes, while PS can only be used in the IMV modes. (see *Basic modes of MV*)⁵.

Many ventilators offer closed-loop modes, achieved through software, and are commonly used in CCUs. These modes track one or more measurements, such as minute ventilation, exhaled carbon dioxide (CO₂), and SpO₂, and automatically make adjustments in certain settings, such as rate, Vt, PS, PEEP, and FIO₂, to achieve a target set in the ventilator settings. The closed-loop approach allows the ventilator's software to make ventilator changes without relying on human action.

Pressure-regulated volume control (PRVC) is an example of a closed-loop ventilation mode. PRVC is a pressure control mode that monitors

Basic modes of MV⁵



Mode	Approach	Description
VC-CMV	Preset tidal volume (Vt)	All breaths are the same, triggered by patient effort or time
PC-CMV	Preset inspiratory pressure	All breaths are the same, triggered by patient effort or time
VC-IMV	Preset Vt plus spontaneous breaths	Mandatory breaths the same, spontaneous breaths vary
PC-IMV	Preset inspiratory pressure	Mandatory breaths the same, spontaneous breaths vary

Examples of ventilator settings orders

Orders for ventilator settings usually have a consistent sequence and use abbreviations presented in this article. Below are two examples of such orders.

Description	Order
Volume-controlled continuous mandatory ventilation with a tidal volume of 500 mL, rate of 10 b/min, FIO ₂ at 50%, plus PEEP at 8 cm H ₂ O	VC-CMV, Vt 500 mL, f-10, FIO ₂ .50, 8 cm H ₂ O PEEP
Pressure-controlled intermittent mandatory ventilation with a PIP of 25 cm H ₂ O, frequency of 8 b/min, 40% oxygen, with the addition of 5 cm H ₂ O PEEP and 10 cm H ₂ O pressure support	PC-IMV, PIP 25, f-14, FIO ₂ .40, 5 cm H ₂ O PEEP, 10 cm H ₂ O PS

the exhaled Vt and adjusts inspiratory pressure to keep the Vt at a targeted level. As a patient's lung compliance improves (less pressure is needed to achieve a given Vt), the ventilator decreases inspiratory pressure to maintain the targeted Vt. Conversely, as the patient's lung compliance decreases, more pressure is needed to reach the targeted Vt, so the ventilator adjusts the inspiratory pressure upward.⁷

Closed-loop modes are helpful but can be problematic in certain patient circumstances. For example, in a patient with high ventilatory demand, such as combined metabolic and respiratory acidosis, issues with uncontrolled pain or anxiety, or a brain injury that causes hyperventilation, the patient's Vt and rate will often be high. If PRVC is used, a set Vt provides feedback to the ventilator to set the amount of inspiratory pressure. This signals the ventilator to reduce the inspiratory pressure, which reduces the amount of support the patient is receiving, despite the higher need for support. If these patients were initially started on PRVC, the mode must be changed to avoid an inordinate WOB.⁷ The healthcare team needs to have a solid understanding of how all ventilator modes work, what changes some modes may make and why, and to closely monitor the patient-ventilator system for possible

conditions that may indicate moving to a different mode or make adjustments to change what the ventilator altered.⁷

Ventilator management

The decision to initiate MV is guided by the patient's clinical status or potential for serious cardiopulmonary deterioration and is based on several factors. For example, there may be issues with poor oxygenation reflected in a low SpO₂ and/or low SaO₂ with low PaO₂ on arterial blood gas (ABG) results. Inadequate ventilation, another reason to use MV, is reflected in high levels of PaCO₂ with accompanying low pH on ABG results.

Patient distress often accompanies issues with oxygenation and ventilation. Signs of patient distress and impending respiratory failure include gasping, labored breathing, use of accessory muscles of respiration, cyanosis of the nail beds or lips, tachycardia, tachypnea, and a look of panic or desperation. Chronic or acute conditions may lead to acute respiratory failure requiring either NIV or MV with artificial airway support. Patient-ventilator system management is guided by ABGs; the patient's clinical status, including vital signs, SpO₂, imaging, and lab results; and monitoring data, such as ventilator pressures and volumes.

Before initiating NIV or MV, a baseline ABG is obtained to help guide the initial settings. In general, most patients begin on VC-CMV.⁸ To address oxygenation issues, a combination of FIO₂ and PEEP will be considered.^{1,8} If the patient has adequate oxygenation (PaO₂ 60-80 mm Hg, SaO₂ >88%) on the baseline ABG, whatever approximate FIO₂ is in place when MV is started will be used for the ventilator setting (see *Examples of ventilator settings orders*).

For example, if the patient has adequate oxygenation on a 40% air-entrainment mask (also known as a Venturi mask), the ventilator will be set at .40 FIO₂ and usually a minimum PEEP of 5 cm H₂O. Conversely, if oxygenation is poor on the initial ABG, the FIO₂ may be set at 1.0 (100% oxygen) with PEEP at 5 cm H₂O or higher. After initiating MV, the oxygen setting is titrated down as the patient's clinical status permits, depending on results from a follow-up ABG obtained about 30 minutes after initiation of MV and based on subsequent monitoring (usually SpO₂). Satisfactory SpO₂ can be used to titrate the FIO₂ if the pulse oximeter reading is tracking correctly and is comparable to the ABG results. Unresponsive oxygenation in the follow-up ABG will generally call for higher FIO₂ if an increase is possible, higher PEEP, or both.

PaCO₂ and pH from the initial ABG guide the settings for rate and Vt in the volume control modes and rate and peak inspiratory pressure (PIP) in the pressure control modes to address ventilation issues. If an acceptable PaCO₂ between 30 and 50 mm Hg is present, the adult rate will be set between 10 and 14 b/min with a Vt set between 6 and 8 mL/kg ideal body weight.^{6,8} If the initial or follow-up ABG reflects a respiratory acidosis (high PaCO₂ and low pH), the rate setting is increased to remove more CO₂. The Vt setting (VC modes) or PIP setting (PC modes) is usually not increased to avoid excessive inspira-

tory pressure and overdistension of the lungs, which can cause barotrauma due to high pressures or volutrauma due to overdistension.

Weaning

Weaning can begin as the need for MV declines, such as when healing from trauma or recovering from surgery. The decision to wean is supported by improving ABG results, chest radiographs, and lab results measuring organ system functioning and improving the patient appearance and vital signs. When weaning and preparing to discontinue MV for adults, the goal is to reach a set FIO_2 of 40% (0.40) or less or a $\text{PaO}_2/\text{FIO}_2$ greater than 150, a PEEP of less than 5 cm H_2O , and pH greater than 7.25. The pH and PaCO_2 should be close to their usual, most recent baseline for patients with COPD.⁸

A spontaneous breathing trial (SBT) is an effective approach to



Patients receiving MV often require close monitoring and intensive nursing and respiratory care.

weaning (see *Assessment for readiness to wean*). For the SBT, the patient is either placed on a T-piece to breathe entirely on their own or, more com-

monly, is changed to the CPAP mode on the ventilator with no rate and a low-level PEEP of 5-8 cm H_2O and possibly low-level PS of 5 cm H_2O . If the results of the SBT are acceptable, the patient should be quickly extubated. However, should the results of the SBT be unacceptable, the patient will be placed back on MV, often with full support from CMV, and allowed to rest. Then, another SBT should be resumed the next day. This process continues daily unless the team decides the patient should not undergo an SBT due to concerns including hemodynamic instability.

The goal is a successful SBT and timely extubation to minimize the time spent receiving MV.⁹ Evidence-based medicine (EBM) is used to guide new approaches to MV and is the reason for using an SBT to determine readiness to extubate. The American College of Critical Care Medicine, the Society of Critical Care Medicine, and the American Society of Health-Systems Pharmacists¹⁰ produced a very useful EBM clinical practice guideline related to managing adult patients in the ICU. This guideline describes the ABCDEF bundle: (A) Assess, prevent, and manage pain; (B) Both spontaneous awakening trials and SBT; (C) Choice of analgesia and sedation; (D) Delirium: assess, prevent, and manage; (E) Early mobility and exercise; and (F) Family engagement and empowerment.¹¹

The ABCDEF bundle performance has returned significant and clinically meaningful improvements in patient outcomes, including survival, MV use, coma and delirium, restraint-free care, ICU readmissions, and post-ICU discharge disposition.¹² With the development of good-quality EBM, patient-ventilator system management will continue to change and evolve.¹³

Ventilator graphics

Most mechanical ventilators in critical care will provide graphic views

Assessment for readiness to wean⁹

A formal assessment is done for readiness to wean, including:

- The cause of respiratory failure (what caused the patient to need MV) should be resolved or significantly improved.
- The patient should have a $\text{PaO}_2/\text{FIO}_2 > 150$, an $\text{FIO}_2 < 0.40$, $\text{PEEP} < 5$ cm H_2O , pH greater than 7.25, rapid shallow breathing index (RSBI) $< 105^*$, a reasonable amount of secretions (not needing frequent suctioning), and improving chest imaging.
- *The formula for RSBI = respiration rate/average tidal volume
- Patients should only need a low level of support to maintain acceptable oxygenation and ventilation, and this should be accomplished by their spontaneous efforts or by noninvasive means.
- The patient should have a strong cough, minimal secretions, and be able to protect and maintain a patent airway.
- The patient should have cardiovascular stability.
 - Adequate management of tachycardia, shock state (low-dose vasopressor support may be acceptable), and no signs of active ischemia or pulmonary edema.
- Glasgow coma scale of 8 or more (lower scores may be acceptable in light of good gag reflex, good cough, and no other contraindications).
- Most patients should be awake, alert, and able to follow commands with no other neurologic issues that impair the ability to breathe spontaneously.
- Hold weaning and SBT if surgery using general anesthesia is planned within the next 24 hours.
- Do not proceed if there is a current use of paralytic agents for any reason.
- An open abdomen and ongoing therapeutic hypothermia are general contraindications.

The patient should have an SBT.

of the patient-ventilator system using scalars and loops. Scalars provide information on each breath based on pressure, flow, or volume over time. These three are often shown simultaneously on one screen. The “sweep speed” (how fast the images move across the screen) can be adjusted to show different time ranges. For example, one could view one or two breaths at a rapid sweep speed or many breaths over a longer time with a slow sweep speed. Loops provide information on the interaction of flow and volume or pressure and volume and show one breath at a time.¹⁴ Understanding and interpreting scalars and loops are very important in troubleshooting issues with the patient-ventilator system.^{3,4,15}

Alarms

MV alarms alert the healthcare team to a problem with the patient ventilator system. Currently, there is no standardization in alarm systems, so alarm packages and characteristics can differ per manufacturer.¹⁶ The main alarms used in MV are for high/low pressure, high/low volume (and minute ventilation), high/low respiratory rate, apnea, and ventilator malfunction. A change in the patient's clinical status may cause more than one alarm to sound. For example, a partially disconnected circuit or major leak may cause alarms for low pressure, low PEEP, low minute volume, or apnea.

Alarms have both a visual and an audible alert, and many have different priority levels (low, medium, high) that have different sounds, increasing frequency, and/or colors. Many alarms automatically reset if the condition resolves itself. Alarms are set on generally accepted parameters that may differ from institution to institution.

Patient care

Patients receiving MV often require close monitoring and intensive nursing



Understanding the basics of MV is foundational to ensuring safe and effective support.

and respiratory care. Nursing care calls for maintaining an effective breathing pattern, adequate and proper gas exchange, nutritional support, avoiding pulmonary infection, and avoiding complications related to immobility. Patient and family education and support are also essential components.

Some patients require frequent suctioning due to excessive secretions; most receive medications for anxiety and pain. Patients should be turned and repositioned at least every 2 hours to avoid pressure injuries related to immobility. ETTs should be changed from one side of the mouth to the other to avoid pressure-induced ulcers at the lip, face, and cheek.¹⁷ Being proactive can reduce healthcare-associated infections, particularly ventilator-associated pneumonia.¹⁸

Communication with patients requiring MV is impaired and may be very difficult or impossible, which may be frustrating for both the patient and the caregiver. Communication and education are also needed with family to help them understand and cope with their loved one's situa-

tion. As mentioned earlier in this article, the ABCDEF ventilator bundle addresses many of these issues and has positively impacted patient outcomes.^{11,12}

Beyond the issues related to MV, some patients may have chest tubes, be recovering from surgery, have traumatic or burn injuries, have dysfunctional cardiac or other organ systems, or be fighting infection. Each comorbidity requires additional care and monitoring, further complicating the provision of quality, safe care.

Conclusion

MV involves many variables to provide a complex support system to the patient. Understanding the basics of MV is foundational to ensuring safe and effective support. As technology advances, ventilators may be more capable of making certain adjustments in the settings to ventilate a patient in the most comfortable, effective way. However, the healthcare team must remain vigilant to oversee changes in the patient's clinical status and ventilator settings to help the patient recover. More experience with the patient-ventilator system helps troubleshoot problems and increases confidence in handling those needing the highest care level. ■

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The author and planners have disclosed no potential conflicts of interest, financial or otherwise.

DOI-10.1097/01.NURSE.0000918996.32229.6b



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